

CHUN-MIN YU[Projects](#) [Research](#) [Contact](#)**NLP • COMPUTER VISION • MULTIMODAL LEARNING**

I'm Chun-Min Yu, a computer science student at National Tsing Hua University. I used to work in IKM Lab, and I'm currently working in NLPSA Lab as Research Assistant. My work spans transformer efficiency, human motion understanding, and robotic vision augmentation.

[Open CV](#)[View selected work](#)

- NTHU B.S. in Computer Science, GPA 3.86/4.3
- Incoming / enrolled M.S. in Computer Science at NTHU
- Focus areas: LLMs, NLP, computer vision, multimodal systems

CURRENT RESEARCH DIRECTION

Multimodal athletic assessment

Developed an end-to-end system for assessing athletic potential through alignment-based analysis of fundamental human movements.

Transformer efficiency

Proposed an attention-skip mechanism that identifies redundant or harmful heads and selectively bypasses them during attention.

LLM Continual Learning Research

Investigating methods to transition LLM in-context learning (ICL) into persistent memory, addressing context window constraints and mitigating catastrophic forgetting.

Projects

01

[NLP](#)[Transformers](#)[Efficiency](#)

Towards Compute-Aware Attention

I proposed an attention skip mechanism for redundant transformer heads. The work studies attention sinks, low-norm value vectors, and head-wise routing behavior to reduce unnecessary computation and improve model performance.

- Observed sink behavior where attention collapses onto low-semantic tokens.
- Designed runtime head masking based on row-wise peakedness.
- Reported gains on GLUE, IMDB, and SQuAD-style tasks.

Achievement

- Awarded 2nd Place in the Computer Science, and 3rd Place in the EECS Undergraduate Project Competitions at NTHU.

[Open paper](#) [Open poster abstract](#) [Watch introduction](#)

02

Multimodal

Human Motion

Evaluation Systems

Talent Profiling via Fundamental Human Movements

This research combines motion encoding, sport-specific language generation, and general evaluation to assess athletic potential from foundational movement patterns. We transform complex video data into "motion tokens", allowing Large Language Models to "read" and understand human movement using their deep, pre-existing sports domain knowledge.

- Encoded motion differences against reference videos with a trainable motion encoder.
- Enable multimodal understanding by mapping multimodal inputs into a form the LLM can reason about using pretrained knowledge without large-scale fine-tuning.
- Provide a unified real-time motion evaluation system that delivers corrective feedback across diverse exercises.

Achievement

- WACV demo session, First author

[Open research poster](#) [Watch demo video](#)

03

Robotic Vision

Generative AI

Video Understanding

Generative Semantic Augmentation for Robot Vision

Robotic vision models often fail under distribution shifts because collecting and labeling real-world videos that cover diverse environments, objects, and interaction dynamics is expensive and risky. We present a practical pipeline for generative semantic augmentation that transforms an input video into new, logically consistent training samples by combining temporally aware object segmentation with mask-constrained video generation.

- Integrated SAM-I2V for stable spatiotemporal segmentation.
- Used Wan 2.1 for latent-space video inpainting and object swapping.
- Added YOLOv8n and Qwen2.5-VL based VideoQA for validation.

[Open full report](#)

EXPERIENCE TIMELINE

2026 - PRESENT

DataLab

Exploring continual learning for LLMs, with an emphasis on context compression, persistent memory, and catastrophic forgetting.

2025 - PRESENT

NLPSA Lab

Working on multimodal learning for athletic assessment, motion analysis, and text-conditioned diffusion-based motion generation.

2024 - 2025

IKM Lab

Studied attention mechanisms in transformers and developed a skip mechanism recognized in NTHU undergraduate competitions.

CAPABILITIES

NLP

LLMs

Computer Vision

Multimodal Learning

PyTorch

HuggingFace

Python

C / C++

Flutter

Firebase

CONTACT

Open to research
collaboration and technical
opportunities.

Reach me at jimmy921018yes@gmail.com, or connect through
[GitHub](#) and [LinkedIn](#).